

$$\textcircled{1} \quad \underline{\underline{6.3 - 14}}$$

$$A^T = -A$$

$$\# \quad \frac{du}{dt} = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} u$$

$$\lambda_1 + \lambda_2 + \lambda_3 = 0$$

$$a) \quad u'_1 = cu_2 - bu_3$$

$$2u_1(cu_2 - bu_3) + 2u_2(au_3 - cu_1) + 2u_3(bu_1 - au_2)$$

$$\Rightarrow \cancel{2c u_1 u_2} - \cancel{2b u_1 u_3} + \cancel{2a u_2 u_3} - \cancel{2c u_1 u_2} + \cancel{2a u_3 u_3} - \cancel{2a u_2 u_3}$$

$$\boxed{= 0}$$

$$\frac{d\|u_t\|^2}{dt} = 0$$

$$\boxed{u_t = u_0}$$

b) $Q = e^{At}$ is orthogonal when A is skew symmetric.

$$e^{A^T t} = I + \frac{A^T}{1!} + \frac{(A^T)^2}{2!} + \dots$$

$$= I - \frac{A}{1!} + \frac{A^2}{2!} - \frac{A^3}{3!} + \dots$$

$$= \underline{\underline{e^{-At}}}$$

② 6.3-24

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$$

$$S \Lambda S^{-1}$$

$$\lambda = 1, 3$$

$$\begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix} \text{ (0)}$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$$

$$S = \begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix}$$

$$S^{-1} = \begin{bmatrix} 1 & -1/2 \\ 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1/2 \\ 0 & 1 \end{bmatrix}$$

$$\# \quad S e^{\Lambda t} S^{-1} = e^{At}$$

$$e^{At} = \begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e^t & 0 \\ 0 & e^{3t} \end{bmatrix} \begin{bmatrix} 1 & -1/2 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} e^t & e^{3t}/2 \\ 0 & e^{3t} \end{bmatrix} \begin{bmatrix} 1 & -1/2 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} e^t & -e^{3t}/2 + e^{3t}/2 \\ 0 & e^{3t} \end{bmatrix}$$

$$= \begin{bmatrix} e^t & e^{3t}/2 [e^{2t} - 1] \\ 0 & e^{3t} \end{bmatrix}$$

③ 6.3-28

$$y'' = -y$$

$$a) \begin{bmatrix} 1 & 0 \\ \Delta t & 1 \end{bmatrix} = B$$

$$B^{-1} = \begin{bmatrix} 1 & 0 \\ -\Delta t & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 \\ -\Delta t & 1 \end{bmatrix} \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \Delta t \\ -\Delta t & (\Delta t)^2 + 1 \end{bmatrix}$$

$$\boxed{\det A = 1}$$

$$b) \det = 1$$

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$$

$$(1-\lambda)(-\lambda) + 1 = 0$$

$$\lambda(\lambda-1) + 1 = 0$$

$$\lambda^2 - \lambda + 1 = 0$$

$$\lambda = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1 \pm \sqrt{3}i}{2}$$

$$\boxed{|\lambda| = 1}$$

$$\downarrow$$

$$e^{i\pi/3}$$

$$\text{so, } (e^{i\pi/3})^6 = \underline{\underline{e^{i\pi}}}$$

④ 6.3-29

$$A = \begin{bmatrix} 1 & \sqrt{2} \\ -\sqrt{2} & -1 \end{bmatrix}$$

$$\lambda_1 + \lambda_2 = 0$$

$$\lambda_1 \lambda_2 = -1 + 2 = 1$$

$$\boxed{\pm i}$$

$$A = \begin{bmatrix} 1 & 2 \\ -2 & -3 \end{bmatrix}$$

$$\lambda_1 \lambda_2 = 1$$

$$\lambda_1 + \lambda_2 = -2$$

$$\boxed{\lambda = -1, -1}$$

⑤ 6-7-20

$$\begin{bmatrix} 1 & -\Delta t/2 \\ \Delta t/2 & 1 \end{bmatrix}$$

$$\begin{array}{c} (\Delta t/2) \\ \hookrightarrow \end{array} \begin{array}{cc|cc} 1 & -\Delta t/2 & 1 & 0 \\ \Delta t/2 & 1 & 0 & 1 \end{array}$$

$$\begin{array}{cc|cc} 1 & -\Delta t/2 & 1 & 0 \\ 0 & \frac{4 + (\Delta t)^2}{4} & -\Delta t/2 & 1 \end{array}$$

$$\begin{array}{c} \hookrightarrow \\ \downarrow \end{array} \begin{array}{cc|cc} 1 & -\Delta t/2 & 1 & 0 \\ 0 & 1 & \frac{-2\Delta t}{4 + (\Delta t)^2} & \frac{4}{4 + (\Delta t)^2} \end{array}$$

$$\begin{bmatrix} \frac{4}{(\Delta t)^2} + 4 & \frac{2\Delta t}{(\Delta t)^2 + 4} \\ \frac{-2\Delta t}{(\Delta t)^2 + 4} & \frac{4}{(\Delta t)^2 + 4} \end{bmatrix}$$

6) 6.4-7

a) $\begin{bmatrix} 1 & b \\ b & 1 \end{bmatrix}$

-ve λ

$$\lambda_1 + \lambda_2 = 2$$

$$\lambda_1 \lambda_2 = 1 - b^2$$

$$\lambda_2 = 3$$

$$\lambda_1 = -1$$

$$-3 = 1 - b^2$$

$$\boxed{b = \pm 2}$$

b) # pivot signs = # negative eigenvalues

c) can't have 2 -ve λ as
 $\text{trace}(A) = \underline{\underline{2}}$

⑦ 6.4 - 16

$$Ax = \lambda x$$

$$x^T Ax = \lambda x^T x$$

$$\lambda = \frac{x^T Ax}{x^T x} \quad \text{Here not}$$

Ans No because $x^T x$ can be zero

⑦ 6.4 - 23

A $\rightarrow$ invertible, orthogonal, permutation,
markov, diagonalizable

B $\rightarrow$ projection, markov, diagonalizable

9) $6.4 - 28$

$$A = \begin{bmatrix} 4 & 2+i \\ 2-i & 0 \end{bmatrix}$$

$$(4-\lambda)(-\lambda) - (2+i)(2-i) = 0$$

$$\lambda^2 - 4\lambda - (4+1)$$

$$(\lambda-5)(\lambda+1) = 0$$

$$\lambda = 5, -1$$

$$\Rightarrow \bar{x}^T A x = \lambda \bar{x}^T x$$

$$Ax = \lambda x$$

$$\bar{x}^T A^T x = \bar{\lambda} \bar{x}^T x$$

$$\begin{aligned} \bar{x}^T A x &= \bar{\lambda} \bar{x}^T x \\ \bar{x}^T A^T x &= \bar{\lambda} \bar{x}^T x \end{aligned}$$

$$\boxed{\lambda = \bar{\lambda}}$$

(10) 6.4.38

$\lambda_{\max}$

$$A = Q \Lambda Q^T$$

$$Q \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_{\max} & \\ & & & \ddots \end{bmatrix} Q^T$$

$$\lambda_1 q_1 q_1^T + \dots$$

$$|q_i| \leq 1$$

$$A = \lambda_1 q_1 q_1^T + \lambda_2 q_2 q_2^T + \dots$$

$$a_{ii} = \sum_{\substack{j=1 \\ j \neq i}}^n q_{ij} \cdot q_{ji} \cdot \lambda_k$$

$$\lambda_k \leq \lambda_{\max}$$

$$|a_{ii}| \leq \lambda_{\max}$$

(11) 8.3 - 9

square of matrix is also matrix

$$\begin{bmatrix} a & c & e \\ b & d & f \\ 1-(a+b) & 1-(c+d) & 1-(e+f) \end{bmatrix}$$

$$a_{11} = a^2 + bc + e(1-(a+b)) =$$

$$a_{21} = ab + bd + f(1-(a+b))$$

$$\begin{aligned} a_{31} &= a(1-(a+b)) + b(1-(c+d)) + (1-(e+f)) \\ &= (1-(a+b))(a-(e+f)+1) + b - bc - ab \\ &= 1-(a+b)(-(e+f)) + \cancel{a} - a^2 - ab + 1 \cancel{a} \cancel{b} \\ &\quad + \cancel{b} - bc - bc \end{aligned}$$

$$a_{11} + a_{21} + a_{31} = 1$$

$$(12) \quad \underline{\underline{8.3 - 12}}$$

$$B = A - I = \begin{bmatrix} -.2 & .3 \\ .2 & -.3 \end{bmatrix}$$

$$\frac{du}{dt} = (A - I)u$$

$A - I$ have $\lambda = 0$
as A have $\lambda = 1$

$$B \rightarrow \lambda = 0, -1/2$$

$-0.2x_1 + 0.3 = 0$
 $x_1 = 3/2$

$$x_1 = \begin{bmatrix} 3/2 \\ 1 \end{bmatrix} \quad x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$u = c_1 e^{\lambda_1 t} x_1 + c_2 e^{\lambda_2 t} x_2$$

$$u = c_1 x_1 + c_2 e^{-t/2} x_2$$

$\rightarrow 0$ as $t \rightarrow \infty$

$$u_{\infty} = c \begin{bmatrix} 3/2 \\ 1 \end{bmatrix}$$

(13) 8.3 - 16

$$A = \begin{bmatrix} .4 & .2 & .3 \\ .2 & .4 & .3 \\ .4 & .4 & .4 \end{bmatrix}$$

$$\lambda = 0, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, 0.2$$

$$\Rightarrow A^k u_0 = S \Lambda^k S^{-1} u_0$$

$$A^k = c_1 \lambda_1^k x_1 + c_2 \lambda_2^k x_2 + c_3 \lambda_3^k x_3$$

$$= c_3 x_3$$

$$\begin{bmatrix} 5.1450 e-01 \\ 5.1450 e-01 \\ 6.8549 e-01 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix}$$